

➤ **Tools for troubleshooting theory (1 hours): overview of tools (Multimeter, POST Card, Software diagnostics).**

Task 1: List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

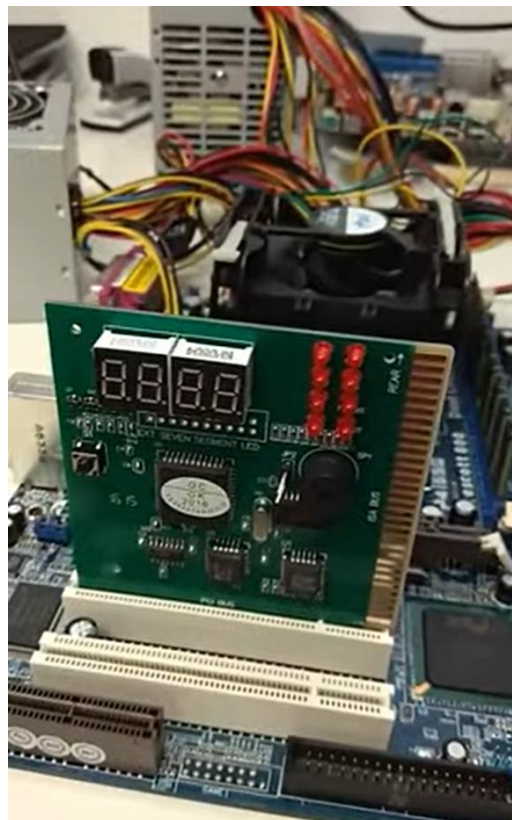
No.	Scenario	How the Multimeter Helps
1	Laptop charger not working	It checks the charger's output voltage to find out whether the adapter is supplying power.
2	Computer power supply problem	It measures voltage at different points to identify abnormal or missing power.
3	Smartphone charging problem	It can be used by a technician to check voltage, continuity, and possible faults in charging-related circuits.



**TASK 2: Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.**

A POST card is a hardware diagnostic tool used during computer startup. It displays hexadecimal POST codes generated by the motherboard BIOS/UEFI. These codes indicate the stage at which the system is starting or where a hardware initialization problem may occur. In the demonstration, the POST code helped identify a hardware-related startup problem.



Caption:

Figure: POST Card displaying diagnostic code

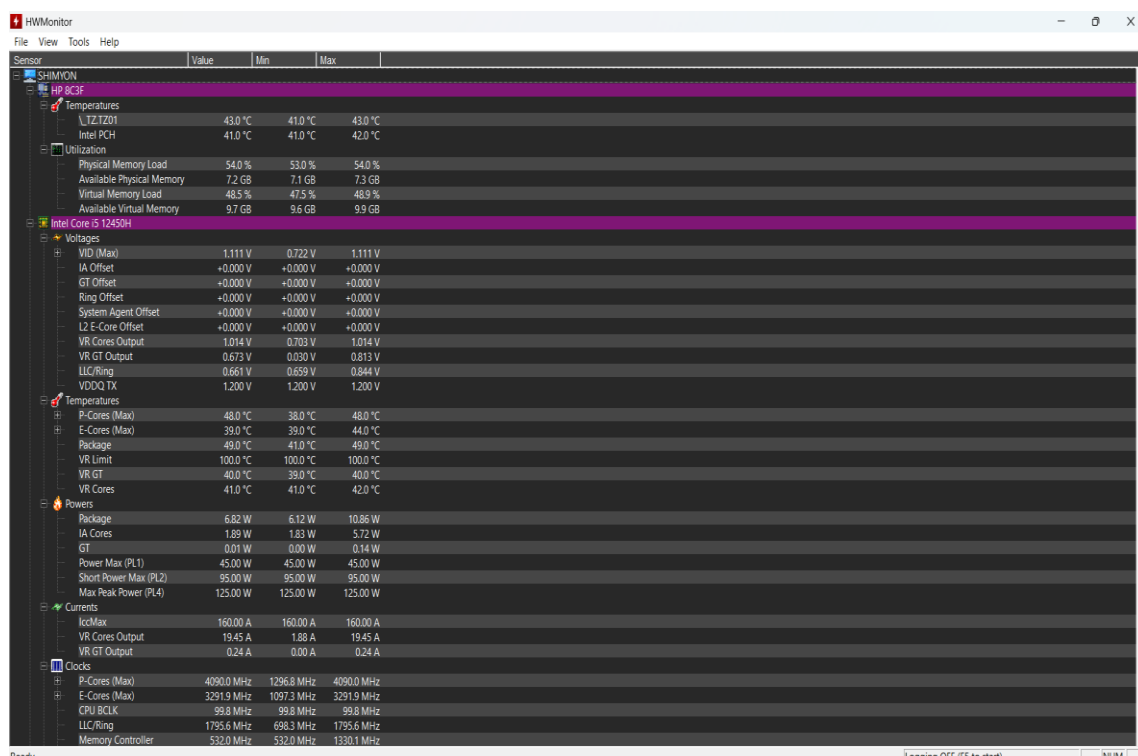
TASK 3: Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.
Constraint: Blur or crop out any personal information before submitting.

Tool Used: HWMonitor

HWMonitor is a free hardware monitoring tool that displays information such as CPU temperature, voltage, fan speed, and system power readings.

Procedure:

1. Open the HWMonitor website.
2. Download the free version.
3. Install HWMonitor on the computer.
4. Open the application.
5. Check the CPU, temperature, voltage, and other system readings.
6. Take a screenshot of the main dashboard.
7. Crop/blur any personal information before adding it to the assignment.



TASK 4: Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, resistance, current and continuity in electrical circuits.	Faulty laptop charger or power supply problem.
POST Card	Checks the computer's startup process and displays POST diagnostic codes.).	RAM, CPU or motherboard initialization problem.
Software Diagnostics	Checks hardware information such as temperature, voltage, fan speed and storage health.	CPU overheating or hard-drive/SSD health problem.

Conclusion:

A multimeter is mainly used for electrical testing, a POST Card is used for hardware startup diagnosis, and software diagnostic tools are used to monitor the health and condition of computer components.